

Vikas Das

Senior Software Engineer, Backend (Java, Spring Boot, Kafka)

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SUMMARY

Backend engineer with 4 years of experience building Java and Spring Boot microservices for 700+ enterprise clients and 500K+ users. Strong in event-driven architecture with Apache Kafka, distributed concurrency, PostgreSQL query optimization, Redis caching and Elasticsearch at 100M+ record scale, and large migrations such as Java 21 across 100+ services.

EXPERIENCE

Senior Software Engineer

Nov 2025 – Present

Peoplehum Pvt Ltd.

Bengaluru, India

- Own end-to-end development of the Attendance, Leave and Performance microservices serving 700+ enterprise clients and 500K+ users, driving system design and architecture decisions.
- Mentored 5 engineers through structured code reviews and design-pattern workshops.
- Migrated the Elasticsearch cluster from v6 to v8 and replaced deep pagination with cursor-based search_after, serving queries across 100M+ records in under 100ms.
- Redesigned attendance calculation from hourly batch jobs to an event-driven architecture, computing 5-6M records daily with sub-second freshness.
- Led the Java 21 upgrade, personally migrating 30+ microservices and establishing patterns and tooling adopted across 100+ services, raising throughput on latency-bound workloads by up to 35%.
- Implemented distributed locking with Redis and Kafka event synchronization for cross-subsidiary employee transfers, keeping payroll, leave and attendance data transactionally consistent.
- Architected a multi-tenant Break Management module enforcing policy across 15+ shift configurations, and launched self-service attendance regularization that handles 5,000+ requests a month without administrators.
- Shipped MCP (Model Context Protocol) servers giving AI agents read-only access to production MariaDB, MongoDB and logs, replacing hand-written queries for support and analytics requests.

Software Engineer

Dec 2024 – Nov 2025

Thermo Fisher Scientific

Bengaluru, India

- Engineered a high-throughput Kafka synchronization microservice that keeps distributed databases consistent, decreasing overall API latency by 30%.
- Lowered database CPU utilization by 50% through PostgreSQL query optimization and targeted Redis caching.
- Provisioned stateful Redis clusters on Kubernetes, cutting monthly QA cloud spend by 40%.
- Configured real-time observability with Prometheus, Grafana and the ELK Stack, reducing mean time to recovery (MTTR) by 45%.

Software Development Engineer I

Dec 2022 – Dec 2024

FlexEHR Healthcare Solutions

Bengaluru, India

- Delivered high-concurrency RESTful microservices processing 10,000+ healthcare transactions daily with 99.9% availability.
- Accelerated PostgreSQL queries by 60% with reworked indexes and execution plans.
- Strengthened HIPAA and SOC 2 compliance across 9 services with OAuth-based API authentication, HashiCorp Vault for secrets, Consul for service discovery and Terraform-managed infrastructure.
- Automated bulk data loading with a CSV ingestion pipeline, eliminating 15 hours of manual data entry weekly, and integrated the WhatsApp Cloud API for appointment notifications, increasing patient engagement by 40%.

Software Engineering Intern

May 2022 – Nov 2022

FlexEHR Healthcare Solutions

Bengaluru, India

- Developed 3 EHR microservices exposing 15+ REST endpoints and tuned MongoDB indexes, shortening document retrieval times by 25%.

PROJECTS

Job Agent | *Python, SQLite, Claude, Playwright, LaTeX* github.com/micro-hawk/job-agent

- Built an open-source job-search agent that discovers roles from Greenhouse, Lever, Ashby and alert emails, ranks matches with a two-stage Claude scorer (Haiku, then Sonnet) under a daily LLM budget, and includes an ATS scorer and LaTeX resume builder; 437 pytest tests.

Webisssh | *Vue, Node.js* github.com/micro-hawk/webisssh

- Designed a self-hosted console that aggregates API-doc URLs from many services into one browser view, organized by environment.

PropT Conrad | *Next.js, Supabase, PostgreSQL* prop-t-conrad.vercel.app

- Created a property management platform with rental listings and a second-hand marketplace for tenants and landlords.

SKILLS

Languages: Java, Python, SQL

Backend: Spring Boot, Spring Security, Hibernate, JPA, Microservices, REST APIs, Distributed Systems, High Availability, System Design, Unit Testing (JUnit, Mockito)

Data & Search: PostgreSQL, Redis, Elasticsearch, MongoDB, MariaDB

Messaging & Concurrency: Apache Kafka, Event-Driven Architecture, Distributed Locking, Virtual Threads, Asynchronous Processing

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, CloudWatch), GCP, Docker, Kubernetes, Terraform, Git, Jenkins, CI/CD

Observability & Security: Prometheus, Grafana, ELK Stack, New Relic, SonarQube, OAuth, HashiCorp Vault, Consul

AI Tooling: Model Context Protocol (MCP), LLM Tool Integration, Context Engineering, Claude

EDUCATION

Savitribai Phule Pune University Aug 2019 – May 2023
Bachelor of Engineering in Information Technology, CGPA 8.35/10 *Pune, India*

Sri Chaitanya 2017 – 2019
Higher Secondary Certificate (HSC), Science *Hyderabad, India*